

balnet: Pathwise Estimation of Covariate Balancing Propensity Scores

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Abstract

We present **balnet**, an R package for scalable pathwise estimation of covariate balancing propensity scores via logistic covariate balancing loss functions. Regularization paths are computed with [Yang and Hastie \(2024\)](#)’s generic elastic net solver, supporting convex losses with non-smooth penalties, as well as group penalties and feature-specific penalty factors. For lasso penalization, **balnet** computes a regularized balance path from the largest observed covariate imbalance to a user-specified fraction of this maximum. We illustrate the method with an application to spatial pixel-level balancing for constructing synthetic control weights for the average treatment effect on the treated, using satellite data on wildfires.

1 Introduction

The propensity score plays a central role in estimating causal effects in observational settings ([Rosenbaum and Rubin, 1983](#)). It is defined as the probability that an individual with a given pre-treatment covariate profile is assigned to the treatment, and is frequently estimated using logistic regression. In high-dimensional settings, such as applications with many confounders, rich basis expansions, or higher-order interactions, a standard approach is to estimate the propensity score using lasso ([Tibshirani, 1996](#)) or elastic net penalized logistic regression ([Zou and Hastie, 2005](#)), as implemented in the widely used R package `glmnet` ([Friedman et al., 2010](#)).

A primary use of the propensity score is inverse probability weighting (IPW), where outcomes are reweighted to construct treated and control groups that are comparable in terms of their pre-treatment covariates. Under the true (oracle) propensity score, IPW exactly balances covariate means between treatment arms and the overall population. However, even without model-misspecification, this exact balancing property does not generally hold for propensity scores estimated using standard maximum likelihood logistic regression. Maximum likelihood estimation prioritizes accurate prediction of treatment assignment, rather than covariate balance. [Imai and Ratkovic \(2014\)](#) introduced the notion of covariate balancing propensity scores, propensity score estimators explicitly designed to optimize covariate balance. Causal effects estimated using these

kinds of weights are more robust to model mis-specification, and more efficient than approaches that estimate propensity scores via maximum likelihood (Ben-Michael et al., 2021; Chattopadhyay et al., 2020).

A number of methods have been proposed to directly target covariate balance. These include the covariate balancing propensity score estimator, CBPS (Fong et al., 2022), which enforces moment conditions via generalized method of moments, and entropy balancing, ebal (Hainmueller, 2012), which directly optimizes weights to achieve balance. Such methods may achieve exact balance in the relatively small datasets commonly encountered in the social sciences. However, in large-scale datasets, these approaches can be computationally challenging to scale and are often infeasible, as exact balance may not be attainable¹. The stable balancing weights approach, sbw (Zubizarreta, 2015), relaxes the requirement of exact balance and instead optimizes weights to satisfy balance levels pre-specified by the researcher. While this approach is effective, it requires the user to choose these levels a priori. If the chosen level is infeasible for a given dataset, or if a different bias–variance tradeoff is desired, the underlying optimization problem must be re-solved. In practice, this leads to iterative tuning between balance constraints and downstream model validation (Keele et al., 2025).

Motivated by these challenges, we seek an algorithmic solution that simplifies the selection of balance levels in a manner analogous to regularization paths for penalized generalized linear models. Our approach builds on recent work that unifies many balance-oriented estimators through covariate balancing loss functions (Zhao, 2019).

For the logistic link, Wager (2024, Chapter 7) note (emphasis added):

...if we believe in a linear-logistic specification and want to use an IPW estimator, then we should learn the propensity model by minimizing the *covariate-balancing loss* function rather than by the usual *maximum likelihood loss* used for logistic regression.

This observation motivates the use of covariate balancing loss functions with modern regularization techniques. The penalized versions of these loss functions correspond to the primal of optimization problems that constrains covariate imbalance (Ben-Michael et al., 2021; Tan, 2020; Zhao, 2019). As a consequence, the interpretation of the penalty parameter differs fundamentally from that in penalized logistic regression, where it acts as a coefficient budget. For lasso in particular, the penalty parameter directly controls the maximum allowable absolute imbalance across covariates, offering an intuitive metric to diagnose overlap issues, as well as monitor solver progress in applications.

To our knowledge, the closest existing software is RCAL (Tan and Sun, 2020), which implements lasso regularization for covariate balancing loss functions. However, because this solver is written entirely in R, it can be difficult to scale to large datasets. This limitation is particularly salient in applications such as spatial or pixel-level balancing, where both the number of observations and covariates are large.

¹A closely related approach from the survey literature, available in the `survey` package (Lumley, 2004), is raking, which calibrates survey weights so that sample marginal distributions match known population margins.

Our package, `balnet` for R (R Core Team, 2024), addresses these limitations by combining covariate balancing loss functions with modern optimization techniques for large-scale regularization problems. Our approach uses proximal quasi-Newton methods for generic convex losses, including algorithmic refinements for coordinate descent such as screening and active-set rules (Friedman et al., 2010; Simon et al., 2013; Tibshirani et al., 2012). The core algorithm builds on the `adelie` C++ elastic net solver by Yang and Hastie (2024), interfaced with R via `Rcpp` (Eddelbuettel and François, 2011), and relies on numerical linear algebra routines from `Eigen` (Guennebaud et al., 2010). In an application from Wu et al. (2023) involving spatial balancing with about 140,000 observations and 500 covariates, `balnet` computes the full regularization path, from raw, unweighted imbalance down to commonly accepted balance thresholds, in under one minute on a standard laptop.

2 Regularization paths for covariate balancing propensity scores

Let $Y_i(1)$ and $Y_i(0)$ denote the potential outcomes under treatment and control, respectively (Imbens and Rubin, 2015). The realized treatment assignment is $W_i \in \{0, 1\}$ and the observed outcome is $Y_i = Y_i(W_i)$, for a sample of units $i = 1, \dots, n$. In observational settings, a common identifying assumption is unconfoundedness: conditional on a set of pre-treatment covariates $X_i \in \mathbb{R}^p$, treatment assignment is as good as random. Under this assumption, a central object is the propensity score,

$$e(x) = \mathbb{P}[W_i = 1 \mid X_i = x].$$

Given suitable overlap conditions, potential outcome means are identified via inverse-propensity weighted (IPW) outcomes. We begin by considering the treated mean, extensions to the average treatment effect (ATE) and the average treatment effect on the treated (ATT) follow by symmetry. The treated mean is identified by

$$\mathbb{E}[Y_i(1)] = \mathbb{E}\left[\frac{W_i Y_i}{e(X_i)}\right].$$

The oracle propensity score also satisfies population-level covariate balance,

$$\mathbb{E}\left[\frac{W_i X_{ij}}{e(X_i)}\right] = \mathbb{E}[X_{ij}], \quad j = 1, \dots, p.$$

For notational simplicity, we describe the setting in terms of linear dependence on X_i , the discussion extends directly to nonlinear specifications via basis expansions and interactions.

Under a logistic propensity score model,

$$e(x) = \frac{1}{1 + \exp(-(\beta_0 + x^\top \beta))}.$$

If β is estimated by maximum likelihood using the standard Bernoulli log-likelihood, the gradient equations are

$$\frac{1}{n} \sum_{i=1}^n \left(W_i - \hat{e}(X_i) \right) X_{ij} = 0, \quad j = 1, \dots, p, \quad (1)$$

where $\hat{e}(X_i)$ are the fitted propensity scores. Maximum likelihood essentially balances predicted and observed treatment. Achieving finite-sample covariate balance using the fitted propensities requires that

$$\frac{1}{n} \sum_{i=1}^n \frac{W_i X_{ij}}{\hat{e}(X_i)} = \frac{1}{n} \sum_{i=1}^n X_{ij}, \quad j = 1, \dots, p. \quad (2)$$

To enforce this covariate balancing condition, we instead work backwards and require the gradients satisfy

$$\frac{1}{n} \sum_{i=1}^n \left(\frac{W_i}{\hat{e}(X_i)} - 1 \right) X_{ij} = 0, \quad j = 1, \dots, p. \quad (3)$$

Substituting in for the logistic propensity scores, this is equivalent to requiring that $\hat{\beta}$ satisfies

$$\frac{1}{n} \sum_{i=1}^n \left(W_i \left[1 + \exp(-(\hat{\beta}_0 + X_i \hat{\beta})) \right] - 1 \right) X_{ij} = 0, \quad j = 1, \dots, p. \quad (4)$$

The loss function that enforces (4) is

$$l_1(\eta) = \frac{1}{n} \sum_{i=1}^n \left(W_i \exp(-\eta_i) + (1 - W_i) \eta_i \right), \quad (5)$$

where $\eta_i = \beta_0 + X_i \beta$. This can be verified by setting the gradient of $l_1(\eta)$ to zero, which yields the balance conditions in (4). The IPW estimator constructed using such weights has favorable statistical properties: when the logistic model is correctly specified it is $\sqrt{n}$ -consistent with efficient variance (Wager, 2024, Theorem 7.1). Intuitively, this approach targets the ratio $W_i/e(X_i)$ (the Riesz representer for $\mathbb{E}[Y_i(1)]$; see e.g. Hirshberg and Wager (2021)), rather than estimating $e(X_i)$ via maximum likelihood and inverting the fitted probabilities².

The balance condition for the control group mean is

$$\frac{1}{n} \sum_{i=1}^n \frac{(1 - W_i) X_{ij}}{1 - \hat{e}(X_i)} = \frac{1}{n} \sum_{i=1}^n X_{ij}, \quad j = 1, \dots, p, \quad (6)$$

which yields an analogous loss $l_0(\eta)$ for estimating propensity scores that target $\mathbb{E}[Y_i(0)]$,

$$l_0(\eta) = \frac{1}{n} \sum_{i=1}^n \left((1 - W_i) \exp(\eta_i) - W_i \eta_i \right). \quad (7)$$

²These weights can also improve doubly robust methods by stabilizing the AIPW debiasing term (Robins et al., 1994); see, for example, Vermeulen and Vansteelandt (2015).

Remark 1. For estimating the ATE, $\mathbb{E}[Y_i(1)] - \mathbb{E}[Y_i(0)]$, this approach requires fitting two propensity score models: one targeting balance for the treated mean and one for the control mean, unlike maximum likelihood which fits one set of coefficients.

Remark 2. For software, it is sufficient to implement only the loss in (5). By symmetry, propensities targeting $\mathbb{E}[Y_i(0)]$ are obtained by solving (5) with inverted treatment indicators $W'_i = 1 - W_i$. The propensities targeting $\mathbb{E}[Y_i(0)]$ can also be used to target the ATT, $\mathbb{E}[Y_i(1) - Y_i(0) \mid W_i = 1]$, since the losses only differ by a scaling (the former weights the control to match $\frac{1}{n} \sum_{i=1}^n X_i$ while the latter weights the controls to match $\frac{1}{\sum_{i=1}^n W_i} \sum_{i=1}^n W_i X_i$).

In practice, directly minimizing (5) may not be feasible. Exact balance may be unattainable in settings with high-dimensional covariates ($p > n$) or limited overlap, potentially producing extreme IPW weights. The covariate balancing loss imposes stronger requirements than maximum likelihood estimation, which only requires overlap, e.g., the absence of complete separation.

Proposition 1 (Proposition S1, Tan (2020)). The loss $l_1(\eta)$ is strictly convex, bounded below, and has a unique minimizer if and only if the following set is empty:

$$\left\{ \beta \neq 0 : \eta_i \geq 0 \text{ if } W_i = 1 \text{ for } i = 1, \dots, n, \text{ and } \frac{1}{n} \sum_{i=1}^n (1 - W_i) \eta_i \leq 0 \right\}.$$

This condition is stricter than logistic separation: it not only forbids a linear hyperplane from perfectly separating treated and controls, but also prevents the weighted control mean from exceeding the treated mean along any linear combination of covariates. Regularization is therefore key to making this approach practical. We consider the penalized objective

$$l_1(\eta) + \lambda P(\beta), \quad (8)$$

where $P(\cdot)$ is a suitable penalty. The default choice in `balnet` is the lasso, $P(\beta) = \sum_{j=1}^p |\beta_j|$. This penalty has a particularly convenient interpretation for balancing losses (Tan, 2020; Zhao, 2019). Exact balance requires

$$\frac{1}{n} \sum_{i=1}^n \frac{W_i X_{ij}}{\hat{e}(X_i)} - \frac{1}{n} \sum_{i=1}^n X_{ij} = 0, \quad j = 1, \dots, p. \quad (9)$$

Under lasso penalization, this constraint is relaxed to a box constraint since the corresponding KKT conditions gives

$$\left| \frac{1}{n} \sum_{i=1}^n \frac{W_i X_{ij}}{\hat{e}(X_i)} - \frac{1}{n} \sum_{i=1}^n X_{ij} \right| \leq \lambda, \quad j = 1, \dots, p. \quad (10)$$

That is, the regularization path indexed by λ yields a sequence of solutions with gradually decreasing maximum imbalance tolerances, corresponding to an ℓ_∞ bound on the covariate imbalance vector.

balnet minimizes (8) using the group elastic net solver of Yang and Hastie (2024), which supports lasso, ridge, elastic net, and group penalties³. Group penalties, optionally combined with penalty factors, allow **balnet** to balance sets of related covariates jointly, such as interactions or categorical expansions, an approach that is useful in many applied settings (e.g., Ben-Michael et al., 2021).

Pathwise optimization is a natural fit for this loss and can provide useful practical diagnostic for overlap. **balnet** constructs a log-spaced λ sequence in the spirit of **glmnet**, starting at the smallest $\lambda^{\max}$ for which the solution has $\hat{\beta} = 0$, and proceeding to a fixed fraction of this value (with default path length 100). For the treated mean, this maximum value is simply given by the largest unweighted treated imbalance,

$$\lambda_1^{\max} = \max_j \left| \frac{1}{n} \sum_{i=1}^n W_i X_{ij} - \frac{1}{n} \sum_{i=1}^n X_{ij} \right|.$$

For lasso penalization, **balnet** also allows users to specify a target $\lambda_{\min}$ corresponding directly to a maximum allowable imbalance. The path solver then attempts to reach this tolerance and gracefully truncates the path if further reductions are infeasible⁴. By default, **balnet** standardizes the covariates, so λ can be interpreted as a limit on the standardized mean difference.

Another benefit of path solvers is the efficient computation of cross-validation grids, as implemented in **cv.balnet**. Direct approaches to cross-validation for regularized models fix a grid of penalty parameters and re-solve the optimization problem at each value (e.g., Chattopadhyay et al. (2020)). In contrast, the path-based approach computes the full sequence of regularization parameters at a computational cost that is often comparable to solving the optimization problem for a single value of λ (Friedman et al., 2007).

3 Application: Pixel-level balancing

We illustrate **balnet** using data from Wu et al. (2023), who estimate the causal effect of past low-intensity fire on future high-intensity fires using NASA’s MODIS instrument (Giglio et al., 2003). Treatment status W_i is equal to 1 for pixels that experienced a low-intensity fire in a given focal year and 0 otherwise, and the outcome is future high-intensity fire status. To construct a control group, the study conducts a synthetic control analysis (Abadie et al., 2010), using covariate balancing to construct ATT weights such that pre-treatment fire history and geographic covariate trajectories are matched.

We focus on a single focal year (2008) and land cover type (conifer forests), yielding a dataset with $n = 141,780$ observations (4,466 treated) and $p = 553$ covariates, measuring past fire history as well as topographic, meteorological, disturbance, and vegetation

³ $l_1(\eta)$ is not a canonical GLM loss due to the multiple W_i terms, this is not an issue as the solver only requires the loss to be convex in the linear predictors η (Yang and Hastie, 2024, Section 4).

⁴If the resulting imbalance remains unsatisfactory, users may consider augmenting the IPW estimator with an outcome model that accounts for the residual imbalance (Athey et al., 2018).

characteristics. The goal is to find weights that match the means of all 553 covariates in the synthetic control group to the corresponding treated-group means. The largest standardized mean difference in the unweighted data is approximately 470, occurring for snow water equivalent in September 2004. We set the parameter `max.imbalance` to 0.05, targeting a maximum allowable standardized covariate imbalance of 0.05. Internally, `balnet` adjusts the λ sequence to range from $\lambda^{\max} = 470$ to $\lambda_{\min} = 0.05$ over `nlambda` = 100 logarithmically spaced values.

Figure 1 summarizes the resulting path fit for a default length of `nlambda` = 100⁵. Motivated by standard diagnostic considerations in IPW analyses (e.g., Austin and Stuart (2015); Greifer (2025); Keele et al. (2025)), some commonly reported metrics are displayed. For a given λ , the estimated propensity scores targeting the control arm are $\hat{e}^{(0)}(X_i; \lambda)$ and the ATT weights are

$$\hat{\gamma}_i(\lambda) = \frac{\hat{e}^{(0)}(X_i; \lambda)}{1 - \hat{e}^{(0)}(X_i; \lambda)}.$$

Let $\bar{X}_{\text{target}}$ and $\bar{\sigma}_{\text{target}}$ denote the treated-group covariate means and standard deviations, and let

$$\bar{X}_0(\lambda) = \frac{\sum_{i:W_i=0} \hat{\gamma}_i(\lambda) X_i}{\sum_{i:W_i=0} \hat{\gamma}_i(\lambda)}$$

denote the weighted control mean. The p -vector of standardized mean differences (SMD) is

$$\text{SMD}(\lambda) = \frac{\bar{X}_0(\lambda) - \bar{X}_{\text{target}}}{\bar{\sigma}_{\text{target}}}.$$

The percentage of bias reduction (PBR) is the reduction in average absolute SMD relative to the unweighted data (corresponding to $\lambda^{\max}$),

$$\text{PBR}(\lambda) = 100 \times \left(1 - \frac{\text{avg}|\text{SMD}(\lambda)|}{\text{avg}|\text{SMD}(\lambda^{\max})|} \right).$$

Finally, the effective sample size (ESS), expressed as a percentage, is

$$\text{ESS}(\lambda) = \frac{100}{n_0} \times \frac{(\sum_{i:W_i=0} \hat{\gamma}_i(\lambda))^2}{\sum_{i:W_i=0} \hat{\gamma}_i(\lambda)^2}, \quad n_0 = \sum_{i=1}^n (1 - W_i).$$

The results show clear improvements in covariate balance, along with an increased concentration of weights on a smaller subset of control units. The right axis in Figure 1 displays the coefficient of variation of the weights $\hat{\gamma}_i(\lambda)$ via the identity linking ESS to the coefficient of variation, $\sqrt{100/\text{ESS}(\lambda) - 1}$.

Figure 2 displays the individual SMD using the estimated weights at $\lambda_{\min}$. Figure 2a shows the ten covariates with the largest unweighted imbalances, all related to snow

⁵As the path is not automatically truncated before reaching `nlambda` steps, further reductions in covariate imbalance would be possible if desired.

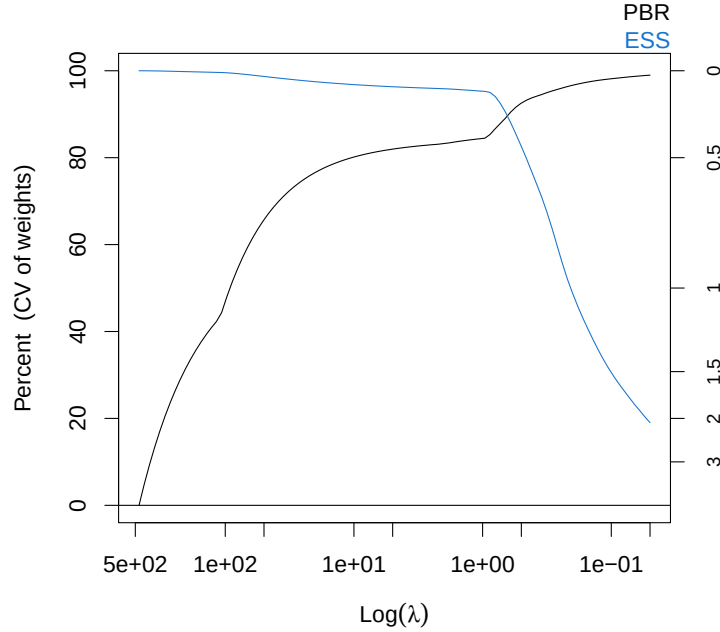
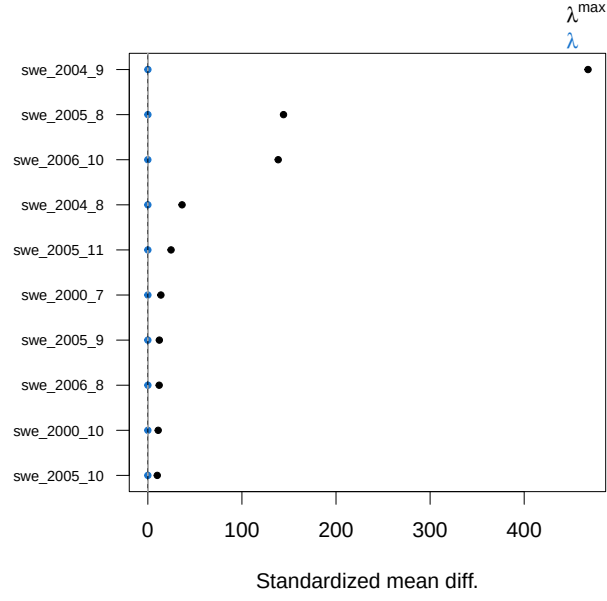


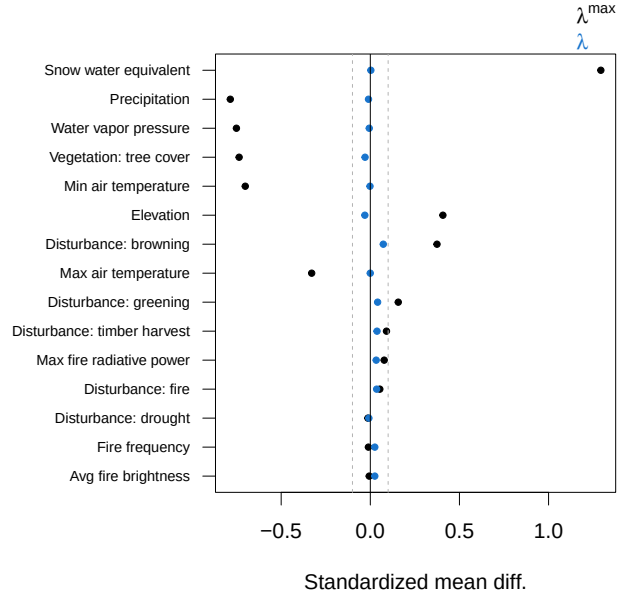
Figure 1: Regularization path for covariate balancing using data from [Wu et al. \(2023\)](#) for treatment year 2008 and conifer forests. The plot shows percentage bias reduction (PBR) and effective sample size (ESS) as functions of λ . The right-hand axis displays the coefficient of variation of the inverse probability weighting (IPW) weights.

water equivalent. Unweighted imbalances are shown as black dots, while weighted imbalances at the selected λ are shown in color. By construction of the covariate balancing loss function, the absolute standardized mean difference for each covariate is bounded above by λ . Visualizing all 500+ covariates in a single panel is impractical, so we supply the `groups` argument to `plot` to aggregate covariates by category. Figure 2b shows the resulting grouped imbalances.

We conclude by examining the estimated propensity scores at $\lambda_{\min}$ and comparing them with those from regularized logistic regression using maximum likelihood. Figure 3 shows the estimated ATT weights for the control units. The treated fraction is only $\frac{4,466}{141,780} \approx 0.03$, so maximum likelihood logistic regression places most propensity estimates near zero in order to reflect the rarity of treatment, yielding ATT weights with point mass near zero. In contrast, the balancing-loss logistic regression assigns more variable weights so the weighted control group more closely matches the treated group.



(a) Standardized mean differences for the 10 most imbalanced covariates.



(b) Standardized mean differences grouped by covariate category.

Figure 2: Standardized mean differences before and after weighting, evaluated at $\lambda_{\min}$ from the path fit, with the unweighted data corresponding to $\lambda^{\max}$, using data from [Wu et al. \(2023\)](#) for treatment year 2008 and conifer forests.

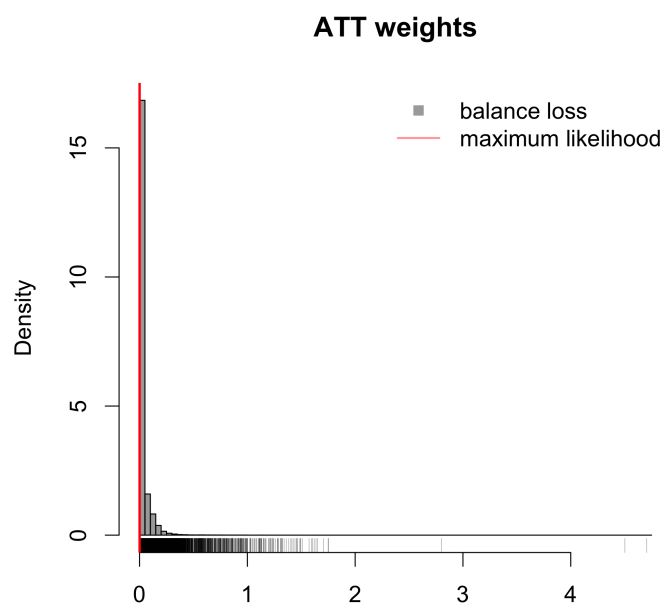


Figure 3: Estimated ATT weights using penalized balance loss at $\lambda_{\min}$ and penalized maximum likelihood loss (default $\lambda_{\min}^{\text{ML}}$).

3.1 Timings

To assess runtime, we conduct a small timing experiment using the data from this section⁶. We define a baseline dataset consisting of the first $n = 100,000$ observations and $p = 500$ covariates from the data described in the previous section. We then double both the sample size and the number of covariates by sampling with replacement from the baseline data. For each setting, we measure the time required for **balnet** to compute a regularization path targeting the ATT with maximum imbalance $\lambda_{\min} \in \{0.05, 0.01\}$. Table 1 reports runtimes obtained on a 8-core laptop with 24 GB of memory, using four cores for solver parallelism. The results suggest that runtime is more sensitive to the λ -sequence endpoints than to increases in the number of observations and covariates.

n	p	max.imbalance ($\lambda_{\min}$)	Runtime (s)	n, p scaling ($\times$)	$\lambda_{\min}$ scaling ($\times$)
100,000	500	0.05	4	—	—
200,000	1,000	0.05	13	3.2	—
400,000	2,000	0.05	53	4.1	—
100,000	500	0.01	43	—	10.8
200,000	1,000	0.01	118	2.7	9.1
400,000	2,000	0.01	385	3.3	7.3

Table 1: Runtimes (in seconds) and relative scaling when doubling both the number of observations and covariates, and tightening balance requirement.

4 Discussion

Regularization is a practical necessity in many modern statistical settings and pathwise solutions are now a standard tool in regularized statistical learning (Friedman et al., 2007, 2010). We use pathwise regularization in a causal setting to target balancing weights for treatment effects under unconfoundedness using logistic covariate balancing loss functions. A natural direction for future work is panel-data settings where separate balancing weights are estimated along both time and unit dimensions, with regularization paths defined accordingly.

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⁶We do not include benchmarks against the R packages mentioned in Section 1, as under our experimental setup they either returned errors or did not complete within a one-hour time window.

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